ZeroPlane: A Mixed-Domain Benchmark and Geometry-Aware Transformer for In-the-wild 3D Plane Reconstruction from a Single Image

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Abstract

Single-image 3D plane reconstruction remains brittle outside the domains on which current systems are trained, in large part because existing methods are developed on narrow indoor or outdoor datasets and are rarely evaluated under a unified zero-shot protocol. We introduce ZeroPlane, which consists of both a mixed-domain benchmark and a geometry-aware Transformer for plane reconstruction. The benchmark consolidates 14 datasets spanning real RGB-D, stereo, and photorealistic synthetic imagery, yielding roughly 569K images and 5.71M plane instances with held-out zero-shot evaluation on NYUv2, 7-Scenes, ApolloScape, and ParallelDomain. The model couples query-based plane detection with decoupled normal and offset prediction, where each quantity is predicted by exemplar classification followed by residual regression, and augments plane queries with auxiliary pixel-wise depth and normal features through geometry-aware attention. Across unseen indoor and outdoor datasets, ZeroPlaneconsistently improves zero-shot geometric reconstruction over prior methods. On NYUv2, mixed-domain ZeroPlaneraises plane recall@0.1m from 14.29 to 21.20 and normal recall@10° from 40.96 to 47.60; on ParallelDomain, it improves plane recall@3m from 35.49 to 46.26 and normal recall@10° from 35.11 to 65.36. High-resolution training further improves high-resolution evaluation on outdoor data and improves normal accuracy indoors. These results suggest that both diverse benchmark coverage and decoupled geometry prediction are necessary for robust in-the-wild planar reconstruction.

1. Introduction

Planes are one of the most useful mid-level geometric primitives for scene understanding, supporting applications such as reconstruction, robotics, autonomous driving, and AR. Considerable progress has been made on recovering planar regions and their 3D parameters from a single image, from early structured predictors and CNN-based models to recent Transformer-based detectors [8, 9, 15, 19, 25]. Yet most existing systems are still trained and validated inside a narrow domain, especially indoor RGB-D benchmarks such as ScanNet or NYUv2 [4, 17], and their behavior under large domain shifts is poorly understood.

This limitation is partly methodological and partly empirical. On the method side, direct plane-parameter regression is sensitive to large cross-domain changes in scene scale, camera intrinsics, and normal distributions. On the benchmark side, existing planar datasets are fragmented across sensors, domains, and annotation conventions. Some provide indoor RGB-D supervision [2, 4, 17], others focus on synthetic driving scenes [1, 5, 14], and others cover only evaluation subsets or adjacent tasks such as depth and semantics [7, 20, 22, 26]. As a result, there is no standard mixed-domain benchmark for unified training and fair zero-shot evaluation of planar reconstruction.

We address both gaps with ZeroPlane. First, we build the ZeroPlane Benchmark, a mixed-domain planar benchmark that aggregates 14 indoor and outdoor datasets into a common reconstruction format. The benchmark contains 10 training/validation sources and 4 held-out evaluation datasets, totaling about 569K images and 5.71M plane instances according to the released statistics in Table 1. Second, we propose a query-based model that explicitly decouples plane normal and plane offset. Instead of regressing the final plane parameter directly, ZeroPlanepredicts normal and offset anchors by classification and then regresses residuals around them. We further inject auxiliary dense depth and normal predictions into plane queries through dedicated geometry-attention branches.

The empirical outcome is strong zero-shot generalization across both indoor and outdoor domains. Mixed-domain training improves over single-dataset training, but the gains are larger when combined with decoupled geometry prediction and geometry-enhanced queries. On NYUv2, 7-Scenes, ParallelDomain, and ApolloScape, ZeroPlaneconsistently improves plane recall and normal recall over PlaneRecTR [15], with especially large gains on unseen outdoor geometry.

Our contributions are:

- We introduce the ZeroPlane Benchmark, a mixed-domain planar reconstruction benchmark spanning 14 datasets, 569K images, and 5.71M plane instances, with a unified

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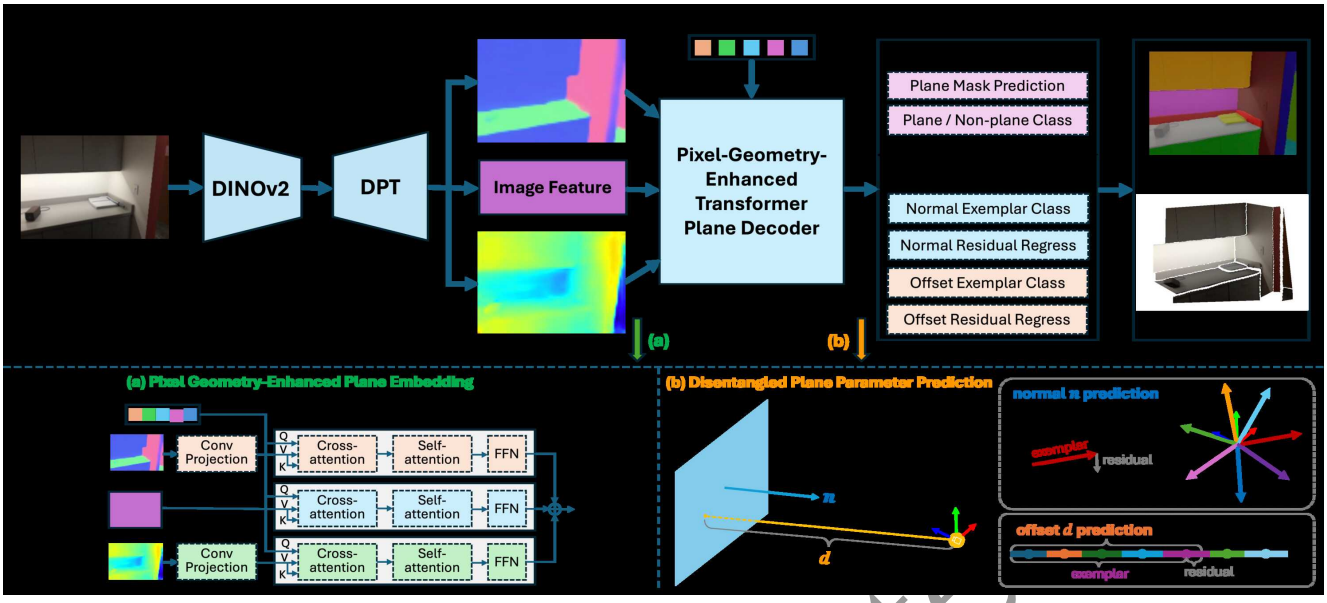

Figure 1. Overview of ZeroPlane. A pretrained visual backbone and dense decoder produce multi-scale features, global pixel depth, and global pixel normals. A Mask2Former-style plane decoder predicts plane masks while separate geometry-attention branches enrich the plane queries with dense depth and normal cues. Plane geometry is decoupled into normal-anchor classification with residual regression and offset-anchor classification with residual regression, after which the final plane parameter is composed from the predicted normal and offset.

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zero-shot evaluation protocol across indoor and outdoor scenes.

- We propose ZeroPlane, a geometry-aware Transformer that decouples plane normal and offset prediction via anchor classification plus residual regression and refines plane queries using auxiliary depth and normal attention.
- We provide extensive zero-shot and ablation experiments showing that both mixed-domain supervision and the proposed geometry formulation materially improve in-the-wild 3D plane reconstruction.

2. Related Work

Single-image plane reconstruction. Plane reconstruction has been studied through structured regression, proposal-based detection, embedding-based grouping, and Transformer architectures. PlaneNet [8] predicts a fixed set of planar primitives from a single RGB image. PlaneRCNN [9] combines proposal-based plane detection with mask refinement. PlaneAE [25] uses associative embeddings to group planar pixels, while PlaneTR [19] and PlaneRecTR [15] move toward query-based reasoning. Our method is closest in spirit to Transformer-based plane detectors, but differs in two ways: it is explicitly designed for cross-domain zero-shot transfer, and it decouples plane normals and offsets instead of treating the plane parameter as a monolithic regression target.

Mixed-domain geometric learning. The importance of dataset mixing for generalization is well established in monocular geometry. Ranftl *et al*. [12] showed that mixing heterogeneous depth datasets substantially improves zero-shot transfer, and DPT [13] demonstrated the value of ViT-based dense prediction. More recent representation learners such as DINOv2 [10] and DUSt3R [23] provide strong geometric priors that transfer across visual domains. ZeroPlanebuilds on this line of work, but targets planar reconstruction specifically, where the output requires coherent instance masks and metric 3D plane parameters rather than only dense depth.

Datasets and benchmarks. Existing data sources cover only slices of the full problem. ScanNet [4], Matterport3D [2], Replica [18], HM3D [24], DIODE [20], and Taskonomy [26] provide indoor geometry; Synthia [14], Virtual KITTI [1, 5], SANPO [22], ParallelDomain [6, 11], and ApolloScape [7] cover outdoor settings; NYUv2 [17] and 7-Scenes [16] remain common indoor testbeds. However, these resources were not assembled as a common planar benchmark with a shared zero-shot protocol. Our benchmark fills that gap.

Table 1. Statistics of the datasets used in our work. Top: Datasets used for training and validation. Bottom: Datasets used for zero-shot evaluation.

Dataset	Indoor	Outdoor	Label	#Training Images	#Validation Images	#Planes
ScanNetv1 [4]	✓		RGB-D	47923	719	281268
ScanNetv2 [4]	✓		RGB-D	140773	859	1445183
Matterport3D [2]	✓		RGB-D	20854	943	227157
Replica [18]	✓		Synthetic	5533	400	90466
HM3D [24]	✓		Synthetic	79994	2000	1298635
DIODE (Indoor Split) [20]	✓		RGB-D	7350	731	37447
Taskonomy [26]	✓		RGB-D	123740	499	1450599
Synthia [14]		✓	Synthetic	24636	999	143698
Virtual KITTI [1, 5]		✓	Synthetic	17720	970	71171
Sanpo (Synthetic Split) [22]		✓	Synthetic	89854	488	647530
NYUv2 [17]	✓		RGB-D	-	654	4849
7-Scenes [16]	✓		RGB-D	-	758	5433
ApolloScape [7] (Stereo Split)		✓	Stereo	-	999	4241
ParallelDomain [6, 11]		✓	Synthetic	-	356	2350

3. The ZeroPlane Benchmark

3.1. Design Goals

The benchmark is designed around three requirements. First, it must span both indoor and outdoor environments, because plane statistics, scale ranges, and camera configurations vary substantially across these domains. Second, it must provide dense plane masks and 3D parameters at a common evaluation interface so that different methods can be compared fairly. Third, it must support zero-shot evaluation on held-out datasets rather than only in-domain testing.

3.2. Data Sources and Scale

Table 1 summarizes the released data. Training and validation use ScanNetv1/v2, Matterport3D, Replica, HM3D, the indoor split of DIODE, Taskonomy, Synthia, Virtual KITTI, and a synthetic split of SANPO [1, 2, 4, 5, 14, 18, 20, 22, 24, 26]. Held-out zero-shot evaluation uses NYUv2, 7-Scenes, ApolloScape, and ParallelDomain [6, 7, 11, 16, 17]. In total, the benchmark covers roughly 569K images and 5.71M plane instances across 14 datasets.

3.3. Construction Pipeline

For each image, we recover planar supervision from available geometry rather than relying on full-scene mesh fitting. Given depth or stereo-derived geometry and camera intrinsics, we back-project pixels to 3D and fit planes at the segment level, which is more robust to incomplete geometry and scales better across heterogeneous datasets. When source datasets provide semantic or instance annotations, they are used to isolate candidate planar regions. Otherwise, panoptic pseudo-labels are generated with a pretrained segmentation model in the style of Mask2Former [3] and used as fitting units.

This per-segment procedure yields dense plane masks and per-plane parameters while avoiding over-merging disconnected regions that happen to be coplanar in world space. We discard ill-conditioned fits and low-planarity segments, standardize camera handling across datasets, and apply domain-balanced sampling to prevent the largest sources from dominating training.

3.4. Tasks and Evaluation Protocol

The benchmark supports three coupled tasks: plane detection, plane segmentation, and 3D plane parameter estimation. For segmentation quality, we report Rand Index (RI), Variation of Information (VOI), and Segmentation Covering (SC), following established planar segmentation practice. For geometry, a predicted plane is matched to a ground-truth plane when their mask IoU exceeds 0.5, after which we measure plane recall under depth and normal thresholds.

The evaluator uses different depth tolerances for indoor and outdoor scenes to reflect their scale regimes. Indoor plane recall is reported at 0.05m, 0.1m, and 0.6m, while outdoor plane recall is reported at 1m, 3m, and 10m. Normal recall is reported at 5°, 10°, and 30° for both domains. This protocol emphasizes both geometric precision and cross-domain robustness.

Beyond aggregate averages, the benchmark is intentionally diagnostic. Indoor scenes emphasize clutter, partial occlusion, and short-range geometry, whereas outdoor scenes stress long-range offsets, wider camera variation, and large support surfaces. Reporting both segmentation and geometry under scale-aware thresholds helps separate grouping failures from parameter-estimation failures, making it harder to hide poor cross-domain transfer behind strong in-domain mask quality alone.

4. Method

4.1. Overview

ZeroPlaneis a query-based planar reconstruction model built on a dense visual backbone, a multi-scale pixel decoder, and a Mask2Former-style Transformer decoder [3]. The released configurations use either DINOv2 with a DPT-style decoder [10, 13] or a DUSt3R encoder-decoder [23]. From an input image, the network predicts plane masks, auxiliary global depth and normals, and 3D plane parameters for a fixed set of plane queries.

4.2. Geometry-Decoupled Plane Prediction

Directly regressing a 3D plane parameter is difficult under cross-domain variation because the normal direction and metric offset have different statistics and different failure modes. We therefore decouple each plane into a unit normal $\mathbf{n}$ and an offset d, and compose the final plane parameter as

$$\mathbf{p} = \frac{\mathbf{n}}{d + \epsilon}. \tag{1}$$

For each query, ZeroPlanepredicts the normal and offset in a classification-then-regression form. Let $\{\mathbf{a}_k^n\}_{k=1}^{K_n}$ be

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normal anchors and $\{a_k^d\}_{k=1}^{K_d}$ be offset anchors, obtained by clustering the mixed-domain training geometry. The model first predicts logits over the anchor sets and then regresses residuals:

$$\hat{c}_n = \arg\max\ f_n(\mathbf{q}), \qquad \hat{c}_d = \arg\max\ f_d(\mathbf{q}), \tag{2}$$

$$\hat{\mathbf{n}} = \mathrm{norm}\big(\mathbf{a}^n_{\hat{c}_n} + \Delta\mathbf{n}_{\hat{c}_n}\big)\,, \tag{3}$$

$$\hat{d} = a^d_{\hat{c}_d} + \Delta d_{\hat{c}_d}, \tag{4}$$

where $\mathbf{q}$ is the plane-query embedding and $\mathrm{norm}(\cdot)$ normalizes to unit length. In the released model, $K_n = 7$ and $K_d = 20$. This decomposition gives the model a coarse geometric prior while retaining the flexibility of residual prediction.

4.3. Geometry-Enhanced Plane Queries

The decoder also predicts global pixel-wise depth and global pixel-wise normals from the dense mask features. These dense predictions serve two purposes. First, they provide auxiliary supervision that stabilizes geometry learning. Second, their embeddings are injected back into the plane queries through dedicated geometry-attention branches.

Concretely, at each decoder stage the plane queries attend not only to the standard multi-scale image features but also to projected dense depth and normal features. The released architecture uses separate attention branches for depth and normal, and their outputs are added to the main query stream. This mechanism gives each plane query direct access to local geometric evidence instead of forcing all geometry reasoning through the RGB feature pathway alone.

4.4. Training Objective

Training follows bipartite matching between predicted queries and ground-truth planes, as in DETR-style instance prediction [3, 21]. The matching cost combines classification and mask terms together with geometry terms. The final loss is a weighted sum of:

- plane classification loss;
- mask binary cross-entropy and Dice losses;
- global pixel depth loss;
- global pixel normal losses using both L1 and cosine penalties;
- normal-anchor classification and residual losses;
- offset-anchor classification and residual losses;
- auxiliary losses at intermediate decoder layers.

The decoder uses 20 plane queries. Low-resolution training uses inputs around 256×192, while the benchmark also supports high-resolution training and evaluation at 640×480.

5. Experiments

5.1. Setup

We evaluate zero-shot transfer on the held-out indoor datasets NYUv2 and 7-Scenes and the held-out outdoor datasets ParallelDomain and ApolloScape. No fine-tuning is performed on these evaluation datasets. We compare against PlaneRCNN [9], PlaneAE [25], PlaneTR [19], and PlaneRecTR [15]. Following the released tables, "S" denotes single-dataset ScanNet training, "S-v2" denotes ScanNetv2 training, and "M" denotes mixed-domain training on the ZeroPlane Benchmark.

5.2. Zero-shot Generalization

Table 2 reports the main comparison. The most consistent gains appear on geometric recall rather than only mask segmentation, which is expected because our main design target is cross-domain 3D geometry.

On NYUv2, mixed-domain ZeroPlaneimproves substantially over PlaneRecTR trained on the same mixed data. With the DUSt3R backbone, plane recall@0.1m improves from 14.29 to **21.20**, plane recall@0.6m from 52.01 to **56.03**, and normal recall@10° from 40.96 to 47.60. VOI also drops from 1.01 to 0.97, while RI and SC remain competitive. On 7-Scenes, ZeroPlaneimproves plane recall@0.1m from 10.97 to **19.14** and normal recall@10° from 26.67 to **36.32**/36.11 depending on the backbone.

The gains are larger outdoors. On ParallelDomain, ZeroPlaneimproves SC from 0.76 to as high as **0.87**, plane recall@3m from 35.49 to **46.26**, and normal recall@10° from 35.11 to **66.13**. On ApolloScape, the DINO-L model achieves the best overall geometry with plane recall@3m rising from 43.06 to **46.40** and normal recall@10° from 33.32 to **40.32**. These results indicate that mixed-domain supervision is important, but the proposed geometry representation contributes additional benefits beyond simple data aggregation.

5.3. Qualitative Results

Figure 2 shows representative zero-shot predictions on all four held-out datasets. The mixed-domain model produces coherent plane partitions across both room-scale indoor scenes and large outdoor layouts, and the reconstructed meshes remain plausible under substantial domain shift. Failure cases still occur on thin structures, reflective surfaces, and far-range outdoor regions, which remain challenging even with mixed-domain training.

6. Analysis

6.1. Component Ablations

Table 3 isolates the main design choices on NYUv2 under both ScanNet-only training and mixed-domain training. Three patterns are consistent.

First, adding the DPT-style dense geometry branch gives a clear gain over the plain query baseline. Under mixed training, recall@0.1m rises from 11.65 to 15.65 and normal recall@10° from 38.15 to 45.23. Second, decoupled

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Table 2. Zero-shot evaluation of different methods or settings on indoor datasets (NYUv2 [17], 7-Scenes [16]) and outdoor datasets (ParallelDomain [6, 11], ApolloScape [7]). (S: trained on ScanNetv1 [4]; S-v2: trained on ScanNetv2, whose training set is much larger than ScanNetv1; M: trained on mixed datasets.)

Evaluation Dataset (Indoor)	Method	Plane Segmentation			Plane Recall (depth)			Plane Recall (normal)		
		RI(↑)	VOI(↓)	SC(↑)	@0.05m	@0.1m	@0.6m	@5°	@10°	@30°
NYUv2 [17]	PlaneRCNN (S-v2) [9]	0.84	1.60	0.61	3.04	9.82	**43.41**	9.24	24.12	43.60
	PlaneAE (S) [25]	0.89	1.40	0.68	1.26	3.36	15.06	11.18	26.54	38.52
	PlaneTR (S) [19]	0.89	1.16	0.71	2.14	6.33	28.30	12.42	27.01	34.63
	PlaneRecTR (S) [15]	**0.92**	1.05	**0.75**	3.3	10.13	41.47	16.42	34.42	45.7
	Ours-DINO-B (S)	**0.92**	**1.02**	**0.75**	**4.33**	**11.03**	41.45	**24.05**	**39.88**	**47.35**
	PlaneRecTR (M) [15]	0.91	1.01	0.73	5.77	14.29	52.01	24.97	40.96	54.71
	Ours-DINO-B (M)	**0.92**	**0.93**	**0.75**	8.54	17.86	55.08	37.29	47.58	57.19
	Ours-DINO-L (M)	**0.92**	0.96	**0.75**	9.32	20.29	55.37	38.15	**47.76**	**56.3**
	Ours-Dust3R (M)	**0.92**	0.97	**0.75**	**9.94**	**21.2**	**56.03**	**38.32**	47.6	56.14
7-Scenes [16]	PlaneRecTR (M) [15]	**0.92**	**1.04**	0.78	3.20	10.97	45.30	11.17	26.67	42.17
	Ours-DINO-B (M)	**0.92**	1.06	**0.79**	8.19	17.19	48.13	23.96	35.43	44.17
	Ours-DINO-L (M)	**0.92**	**1.04**	0.78	7.16	16.25	48.74	25.51	**36.32**	**44.49**
	Ours-Dust3R (M)	**0.92**	1.08	0.78	**9.11**	**19.14**	**49.73**	**26.69**	36.11	44.43
Evaluation Dataset (Outdoor)	**Method**	**Plane Segmentation**			**Plane Recall (depth)**			**Plane Recall (normal)**		
		RI(↑)	VOI(↓)	SC(↑)	@1m	@3m	@10m	@5°	@10°	@30°
ParallelDomain [6, 11]	PlaneRecTR (M) [15]	0.92	0.96	0.76	19.11	35.49	51.92	24.17	35.11	53.36
	Ours-DINO-B (M)	0.94	0.65	0.85	25.96	44.51	63.45	53.23	65.74	69.62
	Ours-DINO-L (M)	**0.95**	**0.57**	**0.87**	**27.66**	44.72	**64.0**	52.47	**66.13**	**70.47**
	Ours-Dust3R (M)	0.94	0.69	0.84	27.4	**46.26**	63.19	**53.91**	65.36	69.87
ApolloScape [7]	PlaneRecTR (M) [15]	0.94	0.38	0.91	10.09	43.06	57.06	22.99	33.32	51.88
	Ours-DINO-B (M)	**0.96**	**0.3**	**0.93**	10.28	40.44	57.34	27.56	39.78	56.05
	Ours-DINO-L (M)	**0.96**	**0.3**	**0.93**	**12.26**	**46.4**	**62.08**	**28.51**	**40.32**	**59.87**
	Ours-Dust3R (M)	0.95	0.35	0.92	10.96	44.21	58.9	24.43	37.3	55.15

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classification-regression for normal and offset improves geometric accuracy further, especially on normal estimation. Under mixed training it raises normal recall@10° from 45.23 to 46.36, and under ScanNet-only training it raises normal recall@5° from 19.78 to **24.27**. Third, geometry-attention gives the strongest full-model configuration overall. With mixed training, the final model reaches **17.86** recall@0.1m and **47.58** normal recall@10°.

The ablation also reveals an important nuance: under ScanNet-only training, the classification-regression head slightly reduces depth recall before geometry-attention is added, even though it improves normal recall. This supports the claim that the decoupled parameterization is most effective when paired with stronger dense geometry cues and more diverse training data.

6.2. Why Mixed Data Matters

Table 4 compares indoor-only, outdoor-only, starter mixed, and full mixed training. Models trained only on one domain transfer poorly to the other: indoor-only training nearly collapses on ParallelDomain, while outdoor-only training nearly collapses on NYUv2. Mixed training substantially improves both directions. The full mixed setup achieves the best balanced performance overall, reaching 17.86 recall@0.1m and 47.58 normal recall@10° on NYUv2 while also reaching **44.51** recall@3m and **65.74** normal recall@10° on ParallelDomain.

Interestingly, domain-specialized training can still win on a few in-domain numbers, such as NYUv2 recall@0.1m for indoor-only training. This makes the benchmark diagnostically useful: it distinguishes specialization from robust transfer rather than conflating them.

Table 3. Ablation studies on the contributed components under both single-dataset (ScanNet) training and mixed-dataset training schemes, evaluated on the NYUv2 dataset.

Training datasets	Component			Plane Recall (depth)			Plane Recall (normal)		
	DPT	Cls-Reg	Geo-Attn	@0.05m	@0.1m	@0.6m	@5°	@10°	@30°
ScanNet	-	-	-	2.76	8.21	39.37	10.87	29.61	45.1
	✓	-	-	4.17	10.66	41.2	19.78	37.93	47.0
	✓	✓	-	3.88	9.71	41.27	**24.27**	38.73	45.84
	✓	✓	✓	**4.33**	**11.03**	**41.45**	24.05	**39.88**	**47.35**
Mixed datasets	-	-	-	3.9	11.65	48.65	20.54	38.15	53.1
	✓	-	-	6.1	15.65	54.18	29.55	45.23	56.59
	✓	✓	-	7.61	16.93	53.97	36.11	46.36	55.39
	✓	✓	✓	**8.54**	**17.86**	**55.08**	**37.29**	**47.58**	**57.19**

The split also reveals complementary failure modes. Indoor transfer is often limited by clutter and fragmented masks around thin or heavily occluded structures, while outdoor transfer is more often limited by metric plane geometry at long range. Evaluating both regimes together therefore tests whether a method has learned a genuinely portable plane representation rather than a domain-specific shortcut.

Table 4. Ablation study on training our framework using different combinations of indoor and outdoor datasets.

Evaluation Dataset	Training Datasets	Plane Recall (depth)			Plane Recall (normal)		
		@0.05m / 1m	@0.1m / 3m	@0.6m / 10m	@5°	@10°	@30°
NYUv2 [17]	Indoor Only	8.41	**18.7**	**55.89**	**37.82**	**47.93**	57.08
	Outdoor Only	0.43	1.71	15.59	7.9	14.68	26.75
	Starter Mixed (110K)	5.24	13.43	46.26	31.2	43.14	51.04
	Full Mixed (560K)	**8.54**	17.86	55.08	37.29	47.58	**57.19**
ParallelDomain [6, 11]	Outdoor Only	**26.72**	43.23	59.87	48.55	59.32	64.43
	Indoor Only	0.81	1.19	3.66	16.94	21.4	37.57
	Starter Mixed (110K)	14.04	35.45	53.53	42.54	55.91	61.36
	Full Mixed (560K)	25.96	**44.51**	**63.45**	**53.23**	**65.74**	**69.62**

6.3. High-resolution Training

Table 5 evaluates training with low-resolution versus high-resolution data and testing against high-resolution annotations. The outcome is mixed but informative. On ParallelDomain, high-resolution training improves every reported metric, including recall@3m from 44.43 to **50.00** and normal recall@10° from 64.34 to **67.91**. On NYUv2, high-resolution training improves VOI, SC, and all normal-recall metrics, but slightly reduces the strict depth-recall thresholds @0.05m and @0.1m. In other words, higher resolution helps boundary quality and normal estimation consistently, while its effect on strict indoor depth recall is more nuanced.

This pattern suggests that extra spatial detail primarily improves orientation and mask localization, while absolute depth alignment remains more sensitive to cross-domain scale variation.

7. Conclusion

We presented ZeroPlane, which combines a mixed-domain planar benchmark with a geometry-aware Transformer for single-image 3D plane reconstruction. The benchmark provides a common training and zero-shot evaluation interface across 14 indoor and outdoor datasets, and the method improves cross-domain generalization by decoupling plane normals and offsets and by injecting dense depth and normal evidence into plane queries. Experiments show clear gains over prior work on held-out indoor and outdoor datasets, especially for geometric recall, while the ablations clarify when mixed data, geometry-attention, and high-resolution training help most. We hope the ZeroPlane Benchmarkcan serve as a stronger community testbed for in-the-wild planar reconstruction beyond narrow single-domain evaluation.

Table 5. Comparison on training using low vs. high resolution data, then evaluation on high-resolution groundtruth.

Evaluation Dataset	Training Resolution	Plane Segmentation			Plane Recall (depth)			Plane Recall (normal)		
		RI(↑)	VOI(↓)	SC(↑)	@0.05m / 1m	@0.1m / 3m	@0.6m / 10m	@5°	@10°	@30°
NYUv2	Low-res	**0.92**	0.93	0.76	**8.35**	**17.78**	55.33	37.41	47.87	57.35
	High-res	**0.92**	**0.84**	**0.77**	7.47	16.19	**57.31**	**40.09**	**50.34**	**59.87**
ParallelDomain	Low-res	**0.94**	0.68	0.84	26.38	44.43	62.68	51.62	64.34	68.43
	High-res	**0.94**	**0.63**	**0.85**	**30.38**	**50.0**	**66.09**	**56.55**	**67.91**	**71.32**

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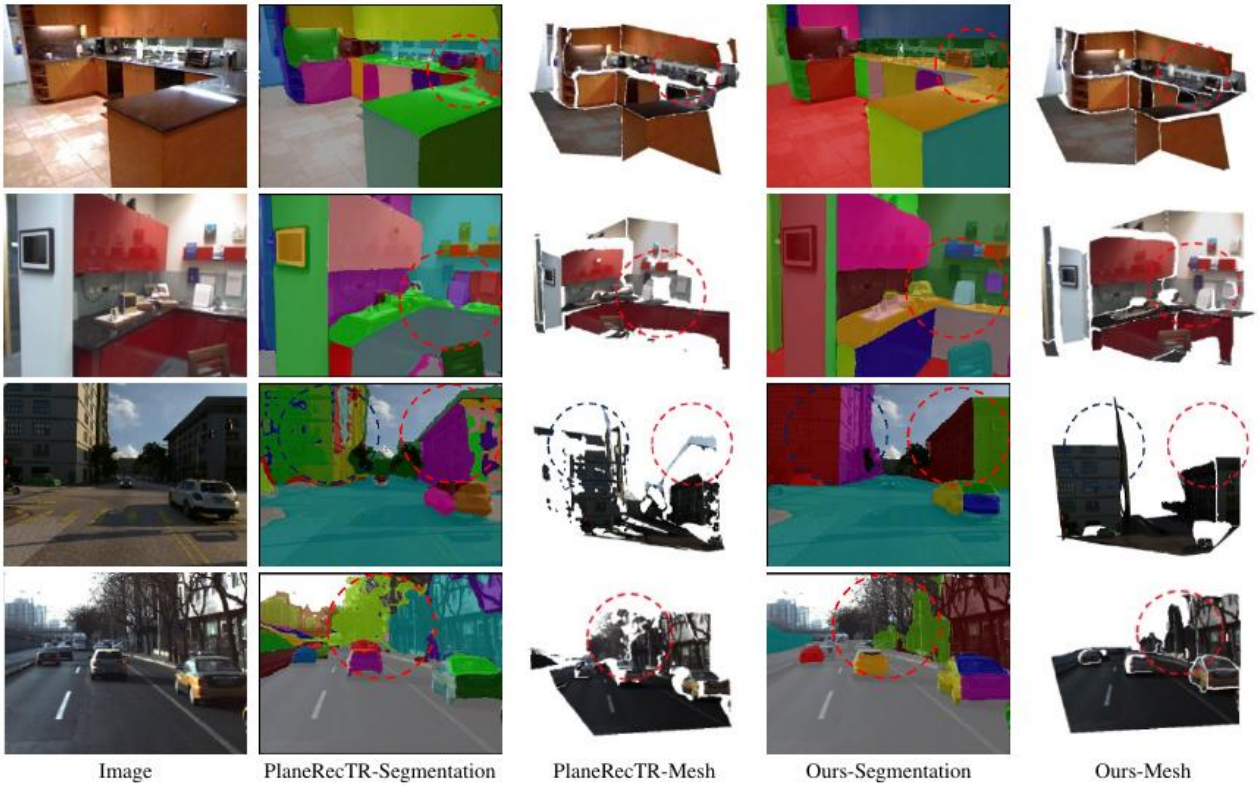

Figure 2. Qualitative zero-shot results of the mixed-domain ZeroPlanemodel on NYUv2, 7-Scenes, ParallelDomain, and ApolloScape. Across both indoor and outdoor scenes, the model produces coherent plane masks and plausible planar meshes despite large changes in appearance, scale, and geometry.

References

[1] Yohann Cabon, Naila Murray, and Martin Humenberger. Virtual kitti 2. *arXiv preprint arXiv:2001.10773*, 2020.

[2] Angel Chang, Angela Dai, Thomas Funkhouser, Maciej Halber, Matthias Niessner, Manolis Savva, Shuran Song, Andy Zeng, and Yinda Zhang. Matterport3d: Learning from rgb-d data in indoor environments. *arXiv preprint arXiv:1709.06158*, 2017.

[3] Bowen Cheng, Ishan Misra, Alexander G Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1290–1299, 2022.

[4] Angela Dai, Angel X Chang, Manolis Savva, Maciej Halber, Thomas Funkhouser, and Matthias Nießner. Scannet: Richly-annotated 3d reconstructions of indoor scenes. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5828–5839, 2017.

[5] Adrien Gaidon, Qiao Wang, Yohann Cabon, and Eleonora Vig. Virtual worlds as proxy for multi-object tracking analysis. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4340–4349, 2016.

[6] Vitor Guizilini, Jie Li, Rareș Ambruș, and Adrien Gaidon. Geometric unsupervised domain adaptation for semantic segmentation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 8537–8547, 2021.

[7] Xinyu Huang, Xinjing Cheng, Qichuan Geng, Binbin Cao, Dingfu Zhou, Peng Wang, Yuanqing Lin, and Ruigang Yang. The apolloscape dataset for autonomous driving. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 954–960, 2018.

[8] Chen Liu, Jimei Yang, Duygu Ceylan, Ersin Yumer, and Yasutaka Furukawa. PlaneNet: Piece-wise planar reconstruction from a single rgb image. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 2579–2588, 2018.

[9] Chen Liu, Kihwan Kim, Jinwei Gu, Yasutaka Furukawa, and Jan Kautz. PlaneRCNN: 3d plane detection and reconstruction from a single image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4450–4459, 2019.

[10] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

[11] Yiming Qian and Yasutaka Furukawa. Learning pairwise inter-plane relations for piecewise planar reconstruction. In *European Conference on Computer Vision*, pages 330–345. Springer, 2020.

[12] René Ranftl, Katrin Lasinger, David Hafner, Konrad Schindler, and Vladlen Koltun. Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer. *IEEE transactions on pattern analysis and machine intelligence*, 44(3):1623–1637, 2020.

[13] René Ranftl, Alexey Bochkovskiy, and Vladlen Koltun. Vision transformers for dense prediction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 12179–12188, 2021.

[14] German Ros, Laura Sellart, Joanna Materzynska, David Vazquez, and Antonio M Lopez. The synthia dataset: A large collection of synthetic images for semantic segmentation of urban scenes. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3234–3243, 2016.

[15] Jingjia Shi, Shuaifeng Zhi, and Kai Xu. Planerectr: Unified query learning for 3d plane recovery from a single view. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9377–9386, 2023.

[16] Jamie Shotton, Ben Glocker, Christopher Zach, Shahram Izadi, Antonio Criminisi, and Andrew Fitzgibbon. Scene coordinate regression forests for camera relocalization in rgb-d images. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2930–2937, 2013.

[17] Nathan Silberman, Derek Hoiem, Pushmeet Kohli, and Rob Fergus. Indoor segmentation and support inference from rgbd images. In *Computer Vision–ECCV 2012: 12th European Conference on Computer Vision, Florence, Italy, October 7-13, 2012, Proceedings, Part V 12*, pages 746–760. Springer, 2012.

[18] Julian Straub, Thomas Whelan, Lingni Ma, Yufan Chen, Erik Wijmans, Simon Green, Jakob J Engel, Raul Mur-Artal, Carl Ren, Shobhit Verma, et al. The replica dataset: A digital replica of indoor spaces. *arXiv preprint arXiv:1906.05797*, 2019.

[19] Bin Tan, Nan Xue, Song Bai, Tianfu Wu, and Gui-Song Xia. PlaneTR: Structure-guided transformers for 3d plane recovery. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4186–4195, 2021.

[20] Igor Vasiljevic, Nick Kolkin, Shanyi Zhang, Ruotian Luo, Haochen Wang, Falcon Z Dai, Andrea F Daniele, Mohammadreza Mostajabi, Steven Basart, Matthew R Walter, et al. Diode: A dense indoor and outdoor depth dataset. *arXiv preprint arXiv:1908.00463*, 2019.

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[21] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

[22] Sagar M Waghmare, Kimberly Wilber, Dave Hawkey, Xuan Yang, Matthew Wilson, Stephanie Debats, Cattalyya Nuengsigkapian, Astuti Sharma, Lars Pandikow, Huisheng Wang, et al. Sanpo: A scene understanding, accessibility, navigation, pathfinding, obstacle avoidance dataset. *arXiv preprint arXiv:2309.12172*, 2023.

[23] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20697–20709, 2024.

[24] Karmesh Yadav, Ram Ramrakhya, Santhosh Kumar Ramakrishnan, Theo Gervet, John Turner, Aaron Gokaslan, Noah Maestre, Angel Xuan Chang, Dhruv Batra, Manolis Savva, et al. Habitat-matterport 3d semantics dataset. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4927–4936, 2023.

[25] Zehao Yu, Jia Zheng, Dongze Lian, Zihan Zhou, and Shenghua Gao. Single-image piece-wise planar 3d reconstruction via associative embedding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1029–1037, 2019.

[26] Amir R Zamir, Alexander Sax, William Shen, Leonidas J Guibas, Jitendra Malik, and Silvio Savarese. Taskonomy: Disentangling task transfer learning. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3712–3722, 2018.

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